VAPO: Value-Model-Based RL for Advanced Reasoning

Achieving State-of-the-Art on AIME 2024

AIME 2024: **60.4** | Base: Qwen2.5-32B | No SFT Data | ByteDance Seed

Surpassing DAPO by **10+ points**

Why Value Models?

Value-model-free methods (GRPO, DAPO) assign trajectory-level advantage uniformly. For long chain-of-thought reasoning, this is insufficient.

Value-Model-Free

GRPO / DAPO

- Uniform advantage across all tokens
- Final reward only
- Token-level errors invisible
- High variance estimates

Value-Model-Based

VAPO

- Token-level credit assignment
- Lower variance estimates
- Generalizes across samples
- Precise error tracing

Three Challenges in Value-Model-Based RL

1. Value Model Bias

Long trajectories and bootstrapped learning compound estimation errors, leading to biased value estimates that degrade policy learning.

2. Heterogeneous Lengths

Responses vary from short answers to 6,000+ token chains. Different lengths require different bias-variance tradeoffs—but one λ is used for all.

VAPO Framework: Integrated Solutions

Value-Pretraining
Pre-trains value model before RL to reduce initial bias

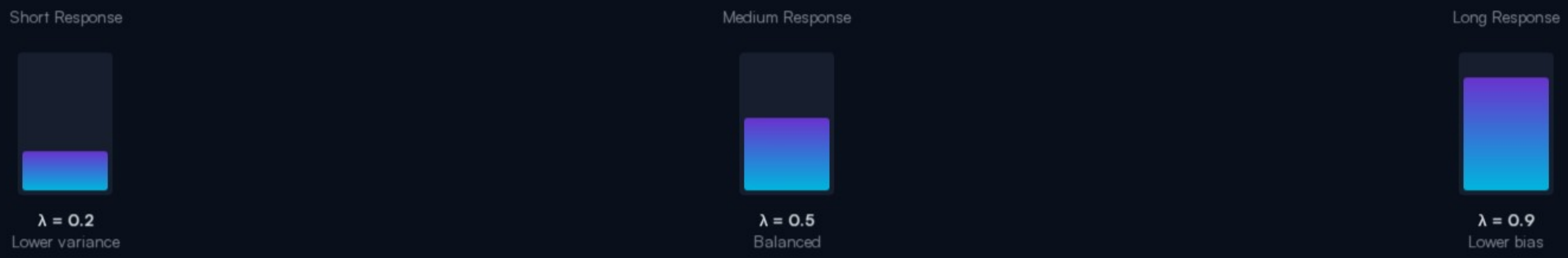
Decoupled-GAE
Separates GAE computation for robust value estimation

Challenge 3
All components improve signal propagation

Supporting Techniques
Clip-Higher (exploration) • Token-level Loss (normalization) • Positive Example LM Loss (self-imitation) • Group-Sampling (baseline estimation)

Length-Adaptive GAE

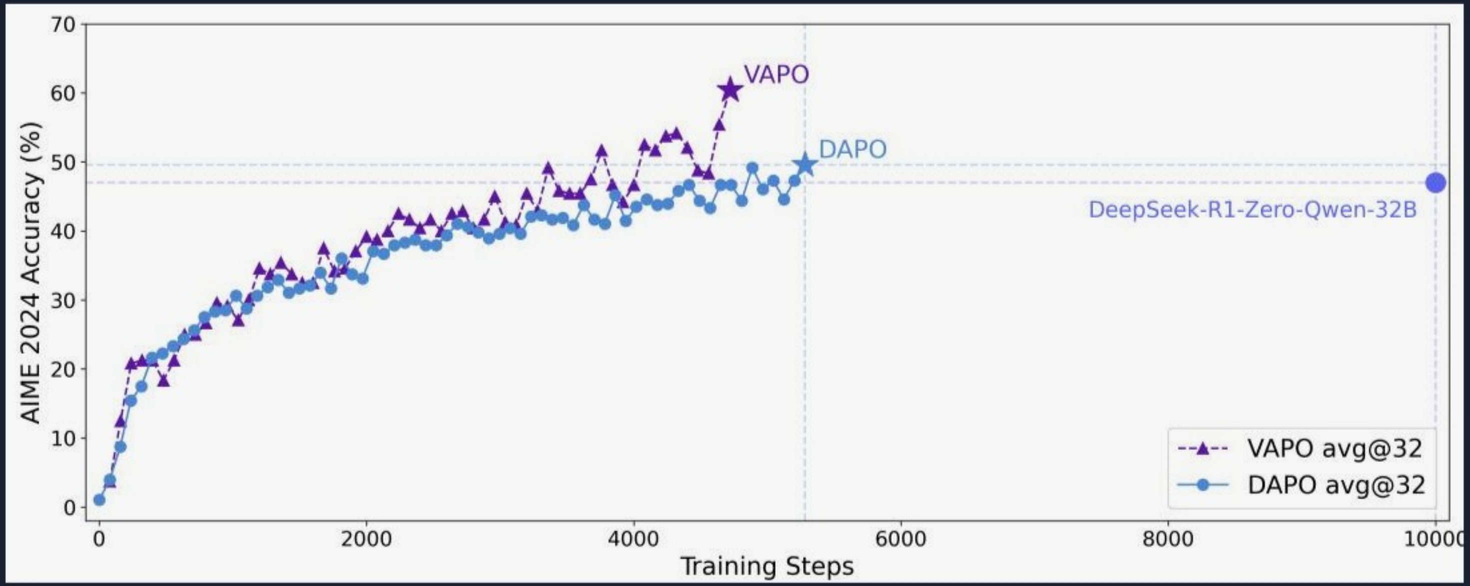
Dynamically adjusts λ based on response length, optimizing the bias-variance tradeoff across heterogeneous sequence lengths.



Key insight: Standard GAE uses fixed λ for all sequences. Length-Adaptive GAE matches λ to sequence characteristics—short sequences don't need high λ (high λ causes high variance when Monte Carlo and bootstrapping disagree). Long sequences benefit from higher λ (more bootstrapping helps when the value function is better trained).

VAPO Scores 60 on AIME 2024

Training Efficiency: SOTA in 5K Steps



VAPO Peak Accuracy
60%
at ~4,500 steps

DAPO Peak
50%
plateaus at ~5,500 steps

VAPO Mean Response
~6,000
tokens (vs ~4,500 for DAPO)

Ablation: Every Component Matters

Component Removed	Score	Relative Performance
Full VAPO (all components)	60	

Key Takeaways

1 **Value-model-based RL can surpass value-free methods**

When the three core challenges (bias, heterogeneous lengths, reward sparsity) are systematically addressed.

2 **Length-Adaptive GAE is critical for long-CoT reasoning**

Dynamic λ adjustment across variable response lengths solves the bias-variance tradeoff that standard GAE cannot.

3 **Value-Pretraining is the single most important component**

Contributing 49 points (11 → 60). Without it, value-model-based RL barely works.